

ACR18787

## Levamisole hydrochloride

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

|                                         |                                                                                                                                                                                                                                                                    |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 产品说明:<br>Product Description:           | 左旋咪唑盐酸盐<br>Levamisole hydrochloride                                                                                                                                                                                                                                |
| Cat No. :<br>Synonyms                   | 187870000; 187870050; 187870100<br>(-)-Tetramisole hydrochloride; L(-)-2,3,5,6-Tetrahydro-6-phenylimidazo[2,1-b]thiazole hydrochloride; (S)-(-)-6-Phenyl-2,3,5,6-tetrahydroimidazo[2,1-b]thiazole hydrochloride                                                    |
| CAS No<br>Molecular Formula             | 16595-80-5<br>C11 H12 N2 S . H Cl                                                                                                                                                                                                                                  |
| Supplier                                | <b>UK entity/business name</b><br>Fisher Scientific UK<br>Bishop Meadow Road,<br>Loughborough, Leicestershire LE11 5RG, United Kingdom<br><br><b>EU entity/business name</b><br>Thermo Fisher Scientific<br>Janssen Pharmaceuticaaan 3a, 2440 Geel, Belgium        |
| Emergency Telephone Number              | For information <b>US</b> call: 001-800-227-6701 / <b>Europe</b> call: +32 14 57 52 11<br>Emergency Number <b>US</b> :001-201-796-7100 / <b>Europe</b> : +32 14 57 52 99<br><b>CHEMTREC</b> Tel. No. <b>US</b> :001-800-424-9300 / <b>Europe</b> :001-703-527-3887 |
| E-mail address                          | begel.sdsdesk@thermofisher.com                                                                                                                                                                                                                                     |
| Recommended Use<br>Uses advised against | Laboratory chemicals.<br>No Information available                                                                                                                                                                                                                  |

### SECTION 2. HAZARD IDENTIFICATION

|                                                                        |                         |                  |
|------------------------------------------------------------------------|-------------------------|------------------|
| Physical State<br>Powder Solid                                         | Appearance<br>Off-white | Odor<br>Odorless |
| <b>Emergency Overview</b><br>Toxic if swallowed. Sensitivity to light. |                         |                  |

#### Classification of the substance or mixture

|                     |            |
|---------------------|------------|
| Acute Oral Toxicity | Category 3 |
|---------------------|------------|

#### Label Elements



Signal Word

Danger

**Hazard Statements**

H301 - Toxic if swallowed

**Precautionary Statements****Prevention**

P264 - Wash face, hands and any exposed skin thoroughly after handling

P270 - Do not eat, drink or smoke when using this product

**Response**

P301 + P310 - IF SWALLOWED: Immediately call a POISON CENTER or doctor

P330 - Rinse mouth

**Storage**

P405 - Store locked up

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

None identified.

**Health Hazards**

Toxic if swallowed.

**Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. Will likely be mobile in the environment due to its water solubility. The product is water soluble, and may spread in water systems.

**Other Hazards**

Toxic to terrestrial vertebrates. This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component                | CAS No     | Weight % |
|--------------------------|------------|----------|
| Levamisole hydrochloride | 16595-80-5 | > 99     |

**SECTION 4. FIRST AID MEASURES****Eye Contact**

Immediate medical attention is required. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.

**Skin Contact**

Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Immediate medical attention is required.

**Inhalation**

Remove from exposure, lie down. Remove to fresh air. If not breathing, give artificial respiration. Immediate medical attention is required.

**Ingestion**

Call a physician immediately. Clean mouth with water.

**Most important symptoms and effects**

No information available.

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES**

**Suitable Extinguishing Media**

Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Chemical foam.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Avoid contact with skin and eyes. Ensure adequate ventilation. Avoid dust formation.

**Environmental Precautions**

See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Do not breathe dust. Do not get in eyes, on skin, or on clothing. Handle product only in closed system or provide appropriate exhaust ventilation.

**Storage**

Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Protect from direct sunlight.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters****Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

**Exposure Controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

## Personal protective equipment

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Nitrile rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Neoprene       | recommendations   |                 |             |                       |
| Natural rubber |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.  
(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particulates filter conforming to EN 143

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Particle filtering: EN149:2001  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                     |                                 |                                          |
|-------------------------------------|---------------------------------|------------------------------------------|
| <b>Appearance</b>                   | Off-white                       |                                          |
| <b>Physical State</b>               | Powder Solid                    |                                          |
| <b>Odor</b>                         | Odorless                        |                                          |
| <b>Odor Threshold</b>               | No data available               |                                          |
| <b>pH</b>                           | 3.0-4.5                         | 5% aq.sol                                |
| <b>Melting Point/Range</b>          | 226 - 231 °C / 438.8 - 447.8 °F |                                          |
| <b>Softening Point</b>              | No data available               |                                          |
| <b>Boiling Point/Range</b>          | No information available        |                                          |
| <b>Flash Point</b>                  | No information available        | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>             | Not applicable                  | Solid                                    |
| <b>Flammability (solid,gas)</b>     | No information available        |                                          |
| <b>Explosion Limits</b>             | No data available               |                                          |
| <b>Vapor Pressure</b>               | No information available        |                                          |
| <b>Vapor Density</b>                | Not applicable                  | Solid                                    |
| <b>Specific Gravity / Density</b>   | No data available               |                                          |
| <b>Bulk Density</b>                 | No data available               |                                          |
| <b>Water Solubility</b>             | 210 g/L (20°C)                  |                                          |
| <b>Solubility in other solvents</b> | No information available        |                                          |

## Levamisole hydrochloride

## Partition Coefficient (n-octanol/water)

Autoignition Temperature 570 °C / 1058 °F

Decomposition Temperature No data available

Viscosity Not applicable Solid

Explosive Properties No information available

Oxidizing Properties No information available

Molecular Formula C11 H12 N2 S . H Cl

Molecular Weight 240.76

## SECTION 10. STABILITY AND REACTIVITY

Stability Stable under normal conditions. Light sensitive.

Hazardous Reactions No information available.

Hazardous Polymerization Hazardous polymerization does not occur.

Conditions to Avoid Exposure to light. Incompatible products.

Materials to avoid Strong oxidizing agents.

Hazardous Decomposition Products Nitrogen oxides (NOx). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Sulfur oxides.  
Hydrogen chloride gas.

## SECTION 11. TOXICOLOGICAL INFORMATION

## Product Information

## (a) acute toxicity;

| Component                | LD50 Oral                | LD50 Dermal | LC50 Inhalation |
|--------------------------|--------------------------|-------------|-----------------|
| Levamisole hydrochloride | LD50 = 180 mg/kg ( Rat ) |             |                 |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

## (d) respiratory or skin sensitization;

Respiratory No data available

Skin No data available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

## Levamisole hydrochloride

|                                            |                                                                |
|--------------------------------------------|----------------------------------------------------------------|
| Target Organs                              | No information available.                                      |
| (j) aspiration hazard;                     | Not applicable<br>Solid                                        |
| Other Adverse Effects                      | The toxicological properties have not been fully investigated. |
| Symptoms / effects, both acute and delayed | No information available                                       |

## SECTION 12. ECOLOGICAL INFORMATION

|                                              |                                                                                                                                                           |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ecotoxicity effects                          | Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants.                               |
| Persistence and Degradability<br>Persistence | Soluble in water, Persistence is unlikely, based on information available.                                                                                |
| Bioaccumulative Potential                    | Bioaccumulation is unlikely                                                                                                                               |
| Mobility in soil                             | The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility Highly mobile in soils |
| Endocrine Disruptor Information              | This product does not contain any known or suspected endocrine disruptors                                                                                 |
| Persistent Organic Pollutant                 | This product does not contain any known or suspected substance                                                                                            |
| Ozone Depletion Potential                    | This product does not contain any known or suspected substance                                                                                            |

## SECTION 13. DISPOSAL CONSIDERATIONS

|                                     |                                                                                                                                                                        |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Waste from Residues/Unused Products | Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations. |
| Contaminated Packaging              | Dispose of this container to hazardous or special waste collection point.                                                                                              |
| Other Information                   | Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.                                          |

## SECTION 14. TRANSPORT INFORMATION

Road and Rail Transport

|                         |                              |
|-------------------------|------------------------------|
| UN-No                   | UN2811                       |
| Proper Shipping Name    | Toxic solid, organic, n.o.s. |
| Technical Shipping Name | Levamisole hydrochloride     |
| Hazard Class            | 6.1                          |
| Packing Group           | III                          |

IMDG/IMO

|                         |                              |
|-------------------------|------------------------------|
| UN-No                   | UN2811                       |
| Proper Shipping Name    | Toxic solid, organic, n.o.s. |
| Technical Shipping Name | Levamisole hydrochloride     |
| Hazard Class            | 6.1                          |
| Packing Group           | III                          |

## SAFETY DATA SHEET

## Levamisole hydrochloride

IATA

UN-No UN2811  
Proper Shipping Name Toxic solid, organic, n.o.s.  
Technical Shipping Name Levamisole hydrochloride  
Hazard Class 6.1  
Packing Group III

Special Precautions for User No special precautions required

## SECTION 15. REGULATORY INFORMATION

## International Inventories

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component                | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|--------------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Levamisole hydrochloride | -                                                   | -                                       | X    | X     | 240-654-6 | -    | X   | X     | X    | X    | X    | KE-33497 |

## National Regulations

## SECTION 16. OTHER INFORMATION

Creation Date 25-Oct-2010  
Revision Date 06-Apr-2024  
Revision Summary SDS sections updated, 2.

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

DNEL - Derived No Effect Level

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

NOEC - No Observed Effect Concentration

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

PNEC - Predicted No Effect Concentration

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

POW - Partition coefficient Octanol:Water

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**PBT** - Persistent, Bioaccumulative, Toxic**vPvB** - very Persistent, very Bioaccumulative**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road**OECD** - Organisation for Economic Co-operation and Development**BCF** - Bioconcentration factor**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code**MARPOL** - International Convention for the Prevention of Pollution from Ships**ATE** - Acute Toxicity Estimate**VOC** - (Volatile Organic Compound)**Key literature references and sources for data**<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**